CSE212 Theory of Computation and Compilers

Parsers

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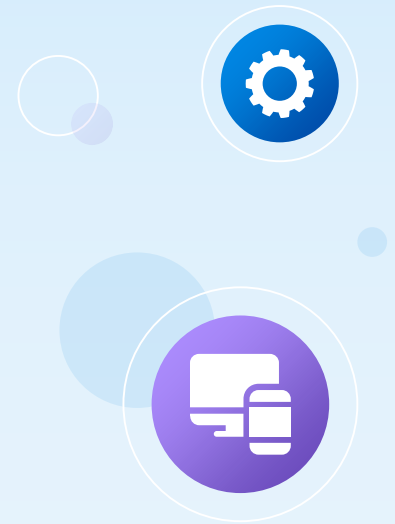
Table of contents



Parsing

- Parsing: Given a stream **s** of words and a grammar **G**, find a derivation in **G** that produces **s**.
- A CFG, G, is a set of rules, or productions, that describe how to form sentences.
- Sentence: a string of symbols that can be derived from the rules of a grammar
- The Context-free Grammers: For a language L, its **CFG** defines the set of strings of symbols that are valid in **L**.
- The collection of sentences that derive from G is called the language defined by **G**, denoted **L(G).**
- The set of languages defined by all possible CFGs is called the set of context-free languages.

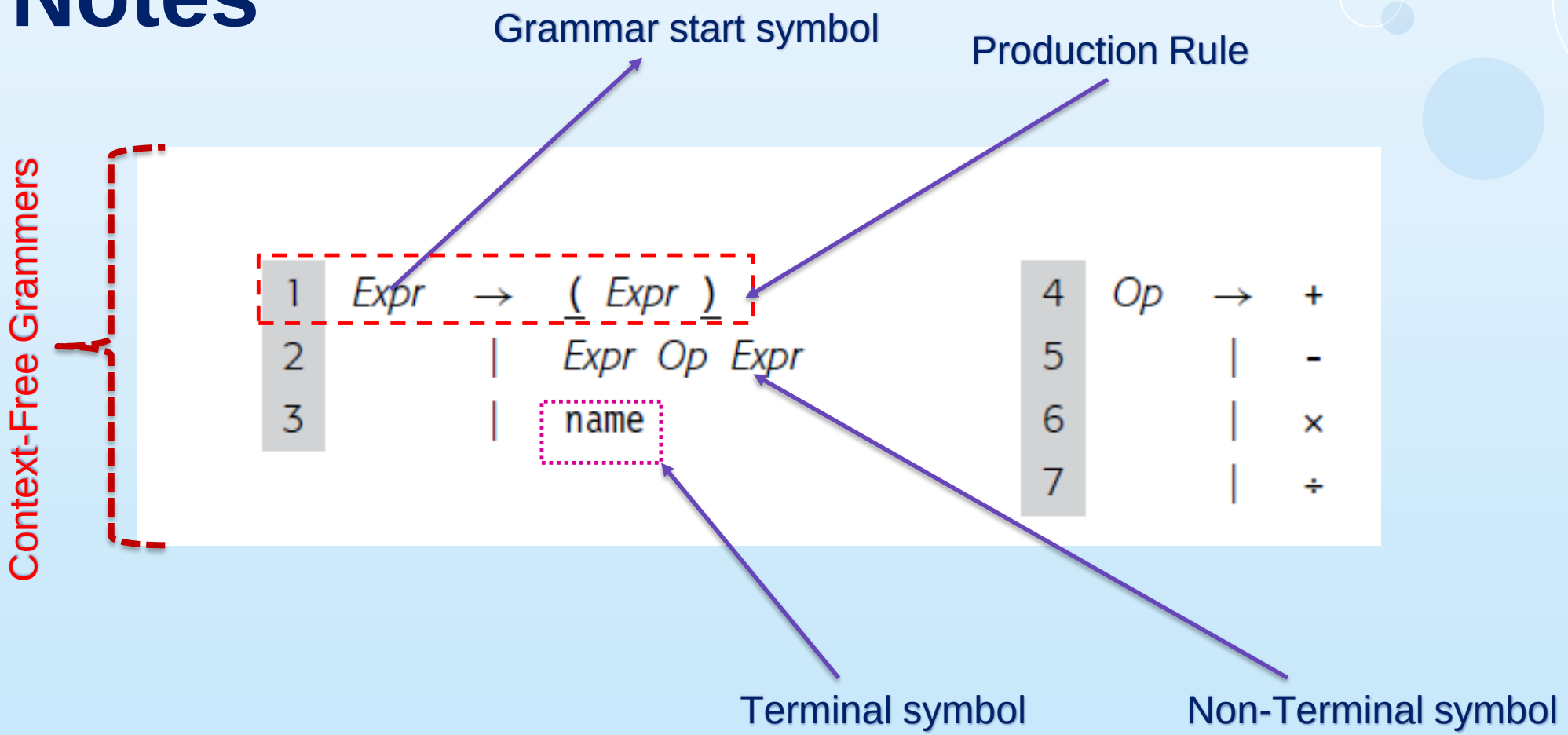
Context-Free Grammars

- Nonterminal symbol: a syntactic variable used in a grammar's productions.
- Terminal symbol: a word that can occur in a sentence.
- Derivation: a sequence of rewriting steps that begins with the grammar's start symbol and ends with a sentence in the language.

1	*Expr*	→	(*Expr*)
2		|	*Expr Op Expr*
3		|	name

4	*Op*	→	+
5		|	-
6		|	×
7		|	÷

Notes


Grammar start symbol
Production Rule
Context-Free Grammers
1 Expr → (Expr)
2 | Expr Op Expr
3 | name
4 Op → +
5 | -
6 | ×
7 | ÷
Terminal symbol
Non-Terminal symbol

Parsing

- Parse tree: a graph that represents a derivation; also called a syntax tree.

- Rightmost derivation: a derivation that rewrites, at each step, the rightmost non-terminal.

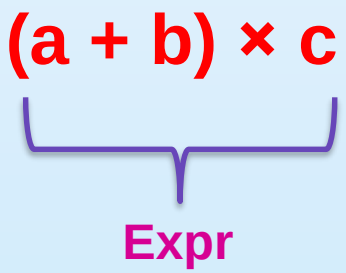
- Leftmost derivation: a derivation that rewrites, at each step, the leftmost nonterminal.

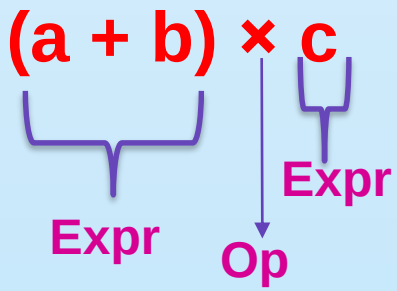
- Ambiguity: A grammar G is ambiguous if some sentence in L(G) has more than one rightmost (or leftmost) derivation.

- A grammar G is ambiguous if some sentence in L(G) has more than one parse tree.

Example: Right-most derivation of (a + b) × c

1	*Expr*	→	(*Expr*)	4	*Op*	→	+
2		|	*Expr Op Expr*	5		|	-
3		|	name	6		|	×
				7		|	÷





Example: Right-most derivation of (a + b) × c

1	*Expr*	→	(*Expr*)	4	*Op*	→	+
2		|	*Expr Op Expr*	5		|	-
3		|	name	6		|	×
				7		|	÷

Rule	Sentential Form
	Expr
2	Expr Op Expr

Left-most NT NT *Right-most NT*

Example: Right-most derivation of (a + b) × c

1	*Expr*	→	(*Expr*)	4	*Op*	→	+
2		|	*Expr Op Expr*	5		|	-
3		|	name	6		|	×
				7		|	÷

Rule	Sentential Form
	Expr
2	Expr Op Expr
3	Expr Op name
6	Expr × name
1	(Expr) × name
2	(Expr Op Expr) × name
3	(Expr Op name) × name
4	(Expr + name) × name
3	(name + name) × name

Example: Right-most derivation of (a + b) × c

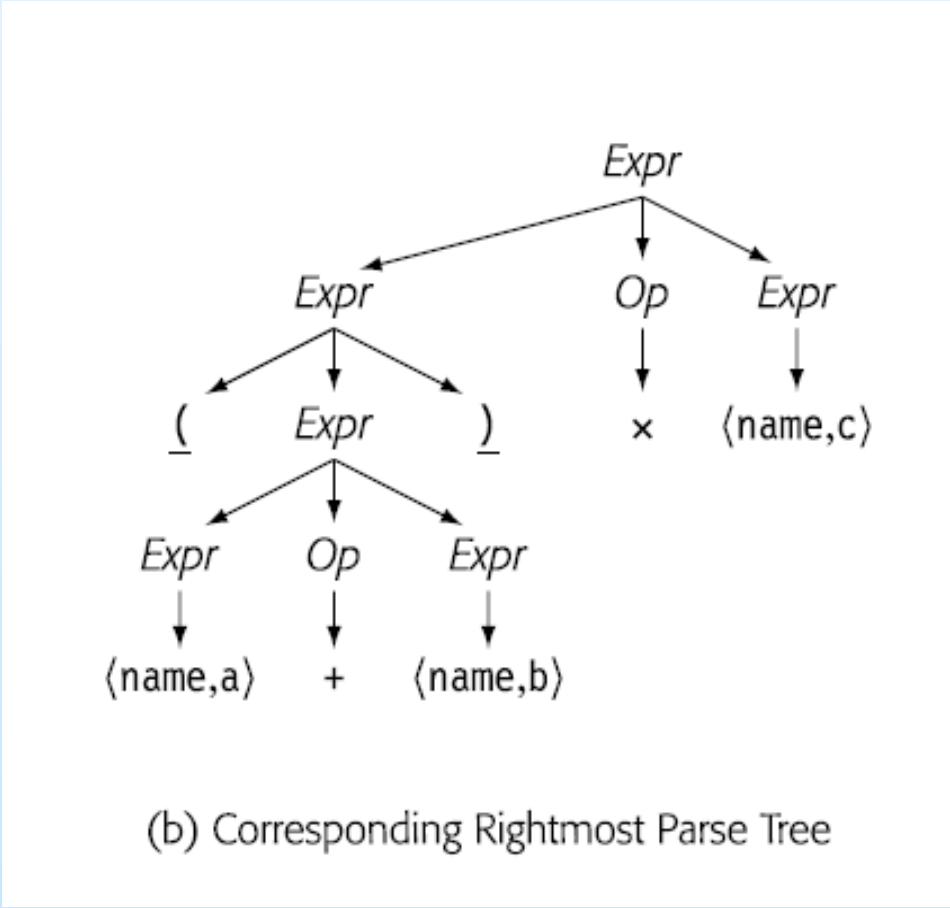
1	*Expr*	→	(*Expr*)	4	*Op*	→	+
2		|	*Expr Op Expr*	5		|	-
3		|	name	6		|	×
				7		|	÷

Grammers or CFG

Rule	Sentential Form
	Expr
2	Expr Op Expr
3	Expr Op name
6	Expr × name
1	(Expr) × name
2	(Expr Op Expr) × name
3	(Expr Op name) × name
4	(Expr + name) × name
3	(name + name) × name

Right-most Derivations

Example: Right-most derivation of (a + b) × c



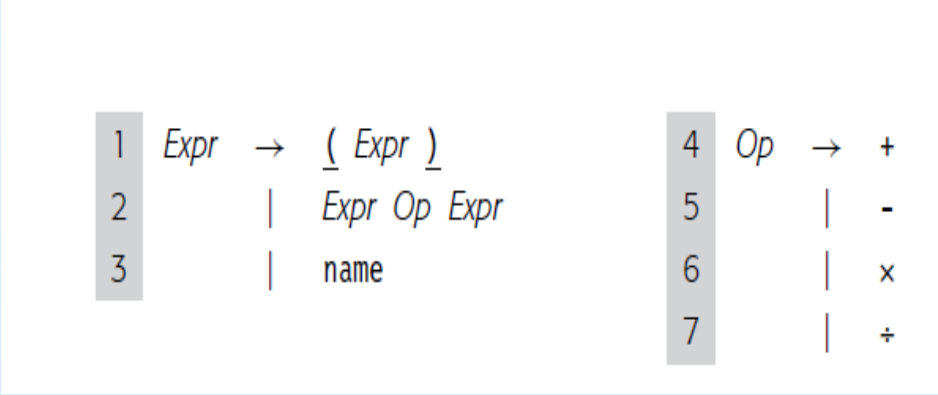
(b) Corresponding Rightmost Parse Tree

Right-most Parse Tree (Syntax Tree)

Rule	Sentential Form
	Expr
2	Expr Op Expr
3	Expr Op name
6	Expr × name
1	(Expr) × name
2	(Expr Op Expr) × name
3	(Expr Op name) × name
4	(Expr + name) × name
3	(name + name) × name

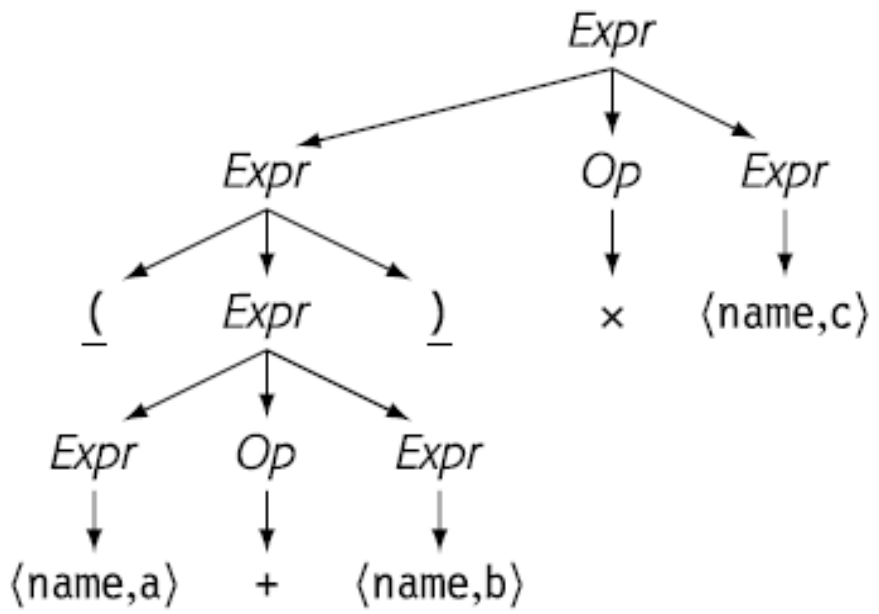
Right-most Derivations

Example: Left-most derivation of (a + b) × c



Rule	Sentential Form
	Expr
2	Expr Op Expr
1	(Expr) Op Expr
2	(Expr Op Expr) Op Expr
3	(name Op Expr) Op Expr
4	(name + Expr) Op Expr
3	(name + name) Op Expr
6	(name + name) × Expr
3	(name + name) × name

Example: Left-most derivation of (a + b) × c



(d) Corresponding Leftmost Parse Tree

Rule	Sentential Form
	Expr
2	Expr Op Expr
1	(Expr) Op Expr
2	(Expr Op Expr) Op Expr
3	(name Op Expr) Op Expr
4	(name + Expr) Op Expr
3	(name + name) Op Expr
6	(name + name) × Expr
3	(name + name) × name

Parsing

- Determine whether the following grammar is suitable for predictive parsing,

$$S \rightarrow (S)|SS|\varepsilon$$

Parsing

- Determine whether the following grammar is suitable for predictive parsing,

$$S \rightarrow (S)|SS|\varepsilon$$

Solution:

- Is this grammar is ambiguous? If yes: it is not suitable
- To prove the grammar ambiguity, Choose a sentence and Do a parse tree derivation

Parsing

- Determine whether the following grammar is suitable for predictive parsing,

$$S \rightarrow (S)|SS|\varepsilon$$

Sentence: ()()()

Parsing

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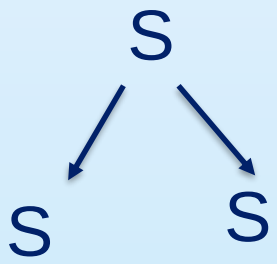

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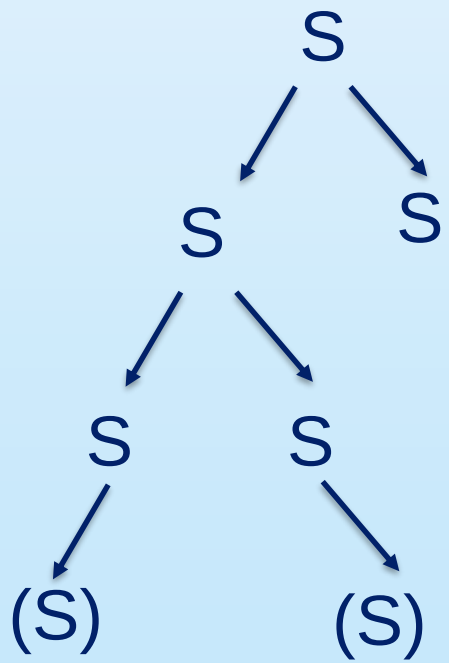

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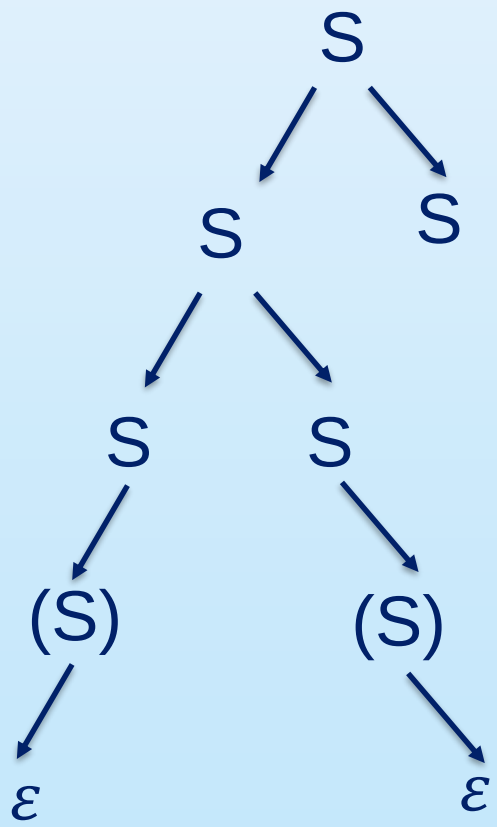

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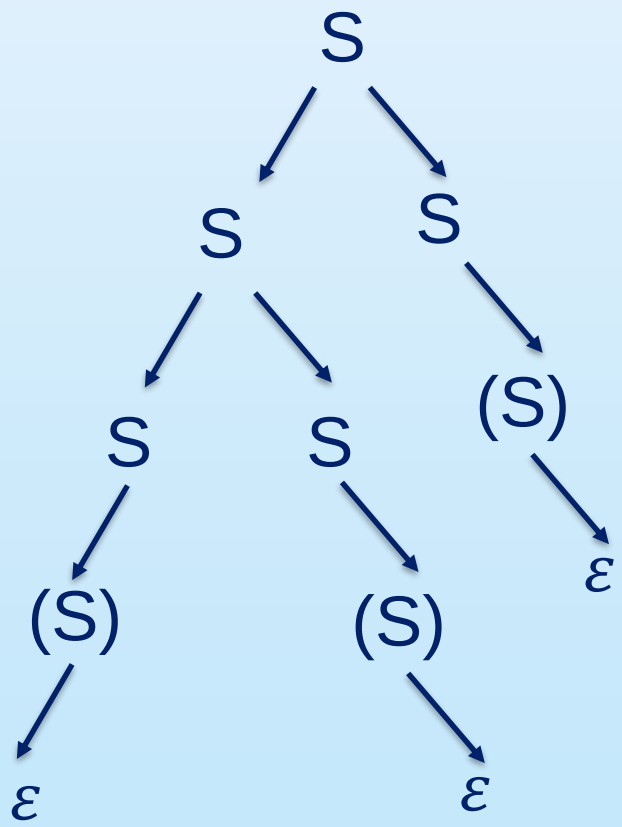

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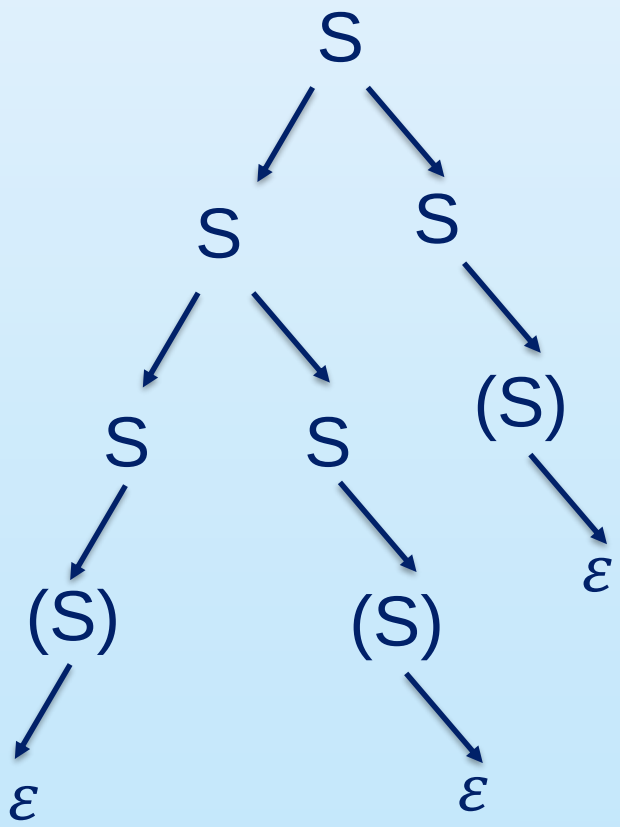

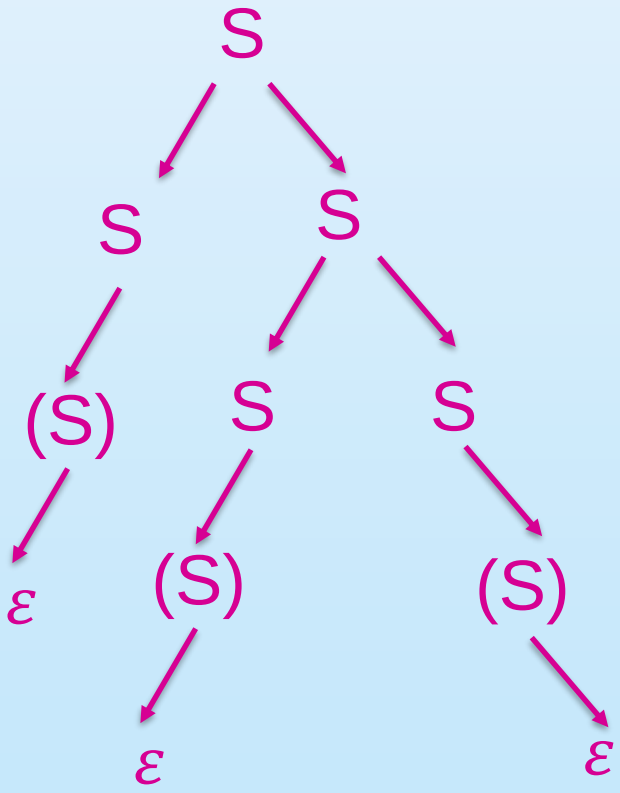

Parsing

- Consider the following two Grammars,
  - G1: S → $SbS|a$
  - G2: S → $aB|a$b, A → $AB|a$, B → $ABb|b$

- Q1: for G2, Define N, T, P, S

- Q2: Which of the following option is correct:
  - Only G1 is ambiguous
  - Only G2 is ambiguous
  - Both G1 and G2 are ambiguous
  - Both G1 and G2 are unambiguous

Parsing

- Consider the following two Grammars,
  - G1: S → $SbS|a$
  - G2: S → $aB|a$b, A → $AB|a$, B → $ABb|b$

- Q1: for G2, Define N, T, P, S
  - **Non-terminals N:** S, A, B
  - **Terminals T:** a, b
  - **Start Symbols S:** S
  - **Production Rules P:** S → $aB|a$b, A → $AB|a$, B → $ABb|b$

Parsing

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- Q2: Which of the following option is correct:
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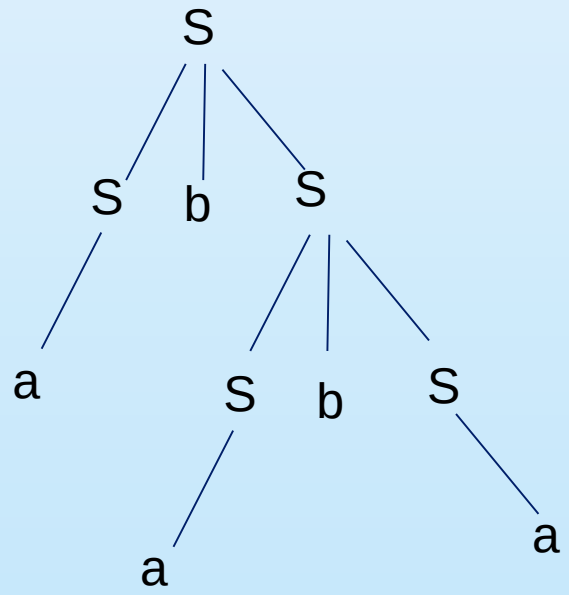
- Consider the following two Grammars,
  - G1: S → $SbS|a$

Consider a sentence: ababa



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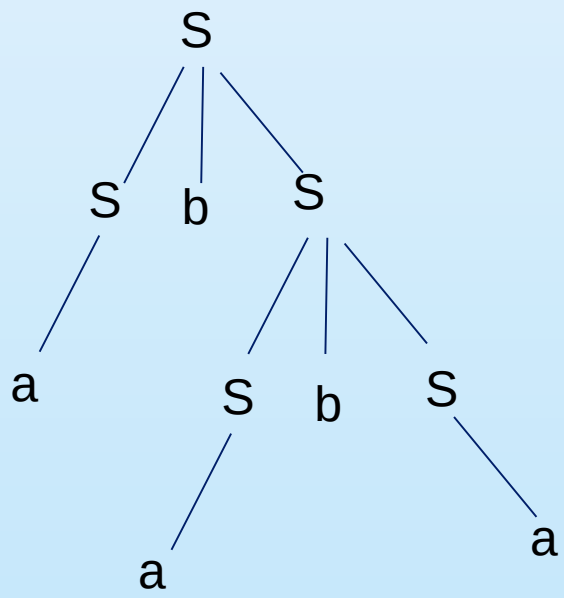
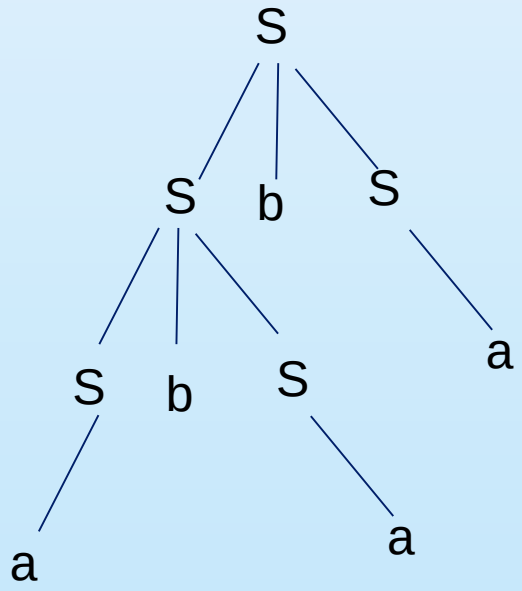
  - G1: S → $SbS|a$

Consider a sentence: ababa

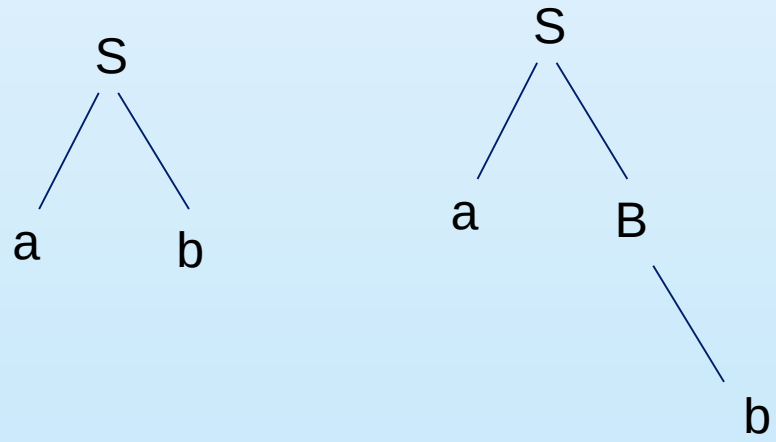




Parsing

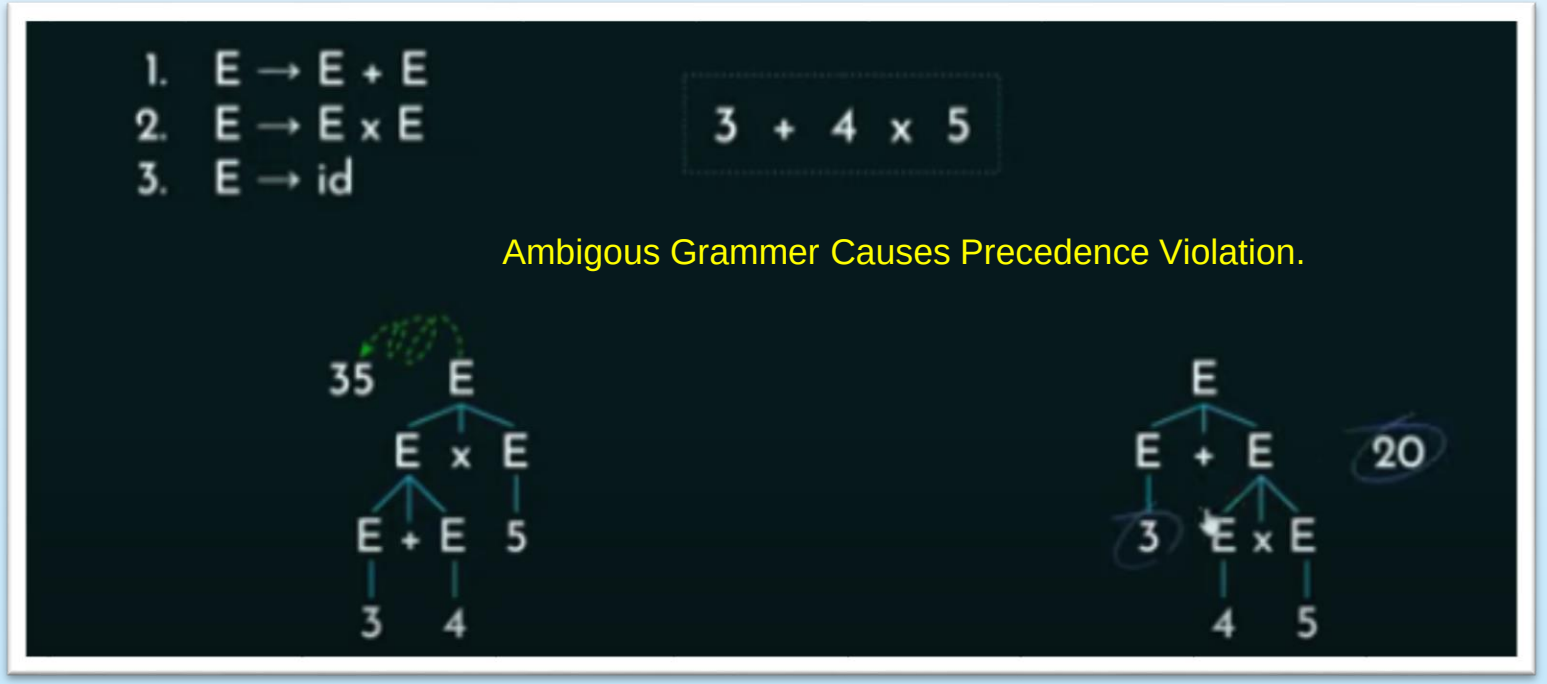
- Consider the following two Grammars,
  - G2: S → $aB|a$b, A → $AB|a$, B → $ABb|b$

Consider a sentence: ab



Ambiguity

- Ambiguity Problems:
  - Precedence Property Violation
  - Associatively Property Violation



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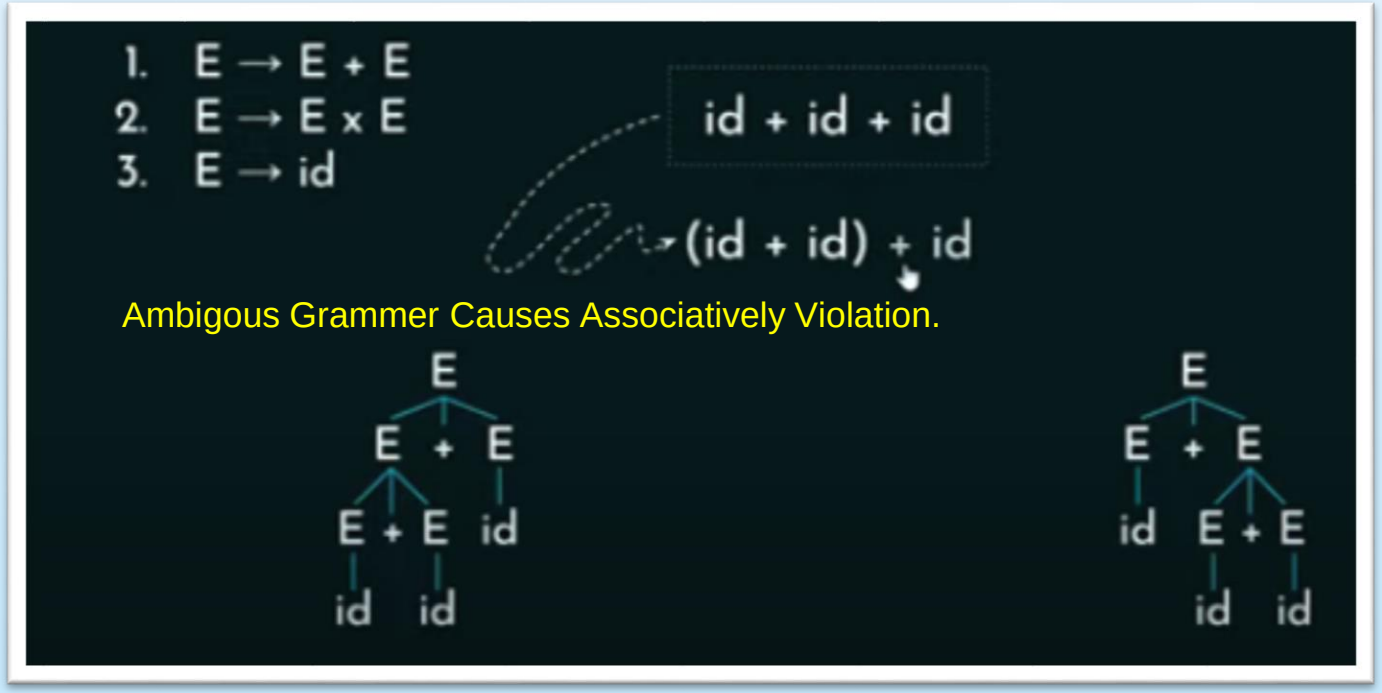

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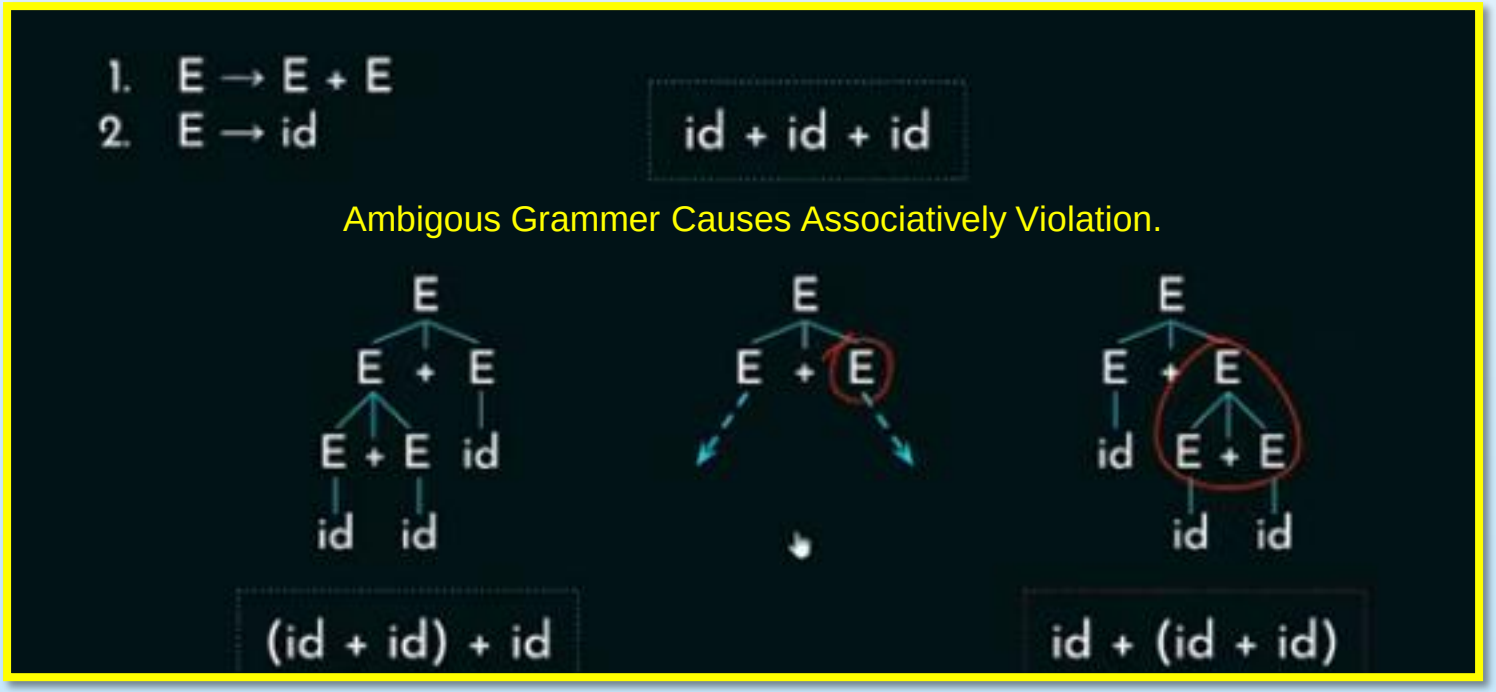
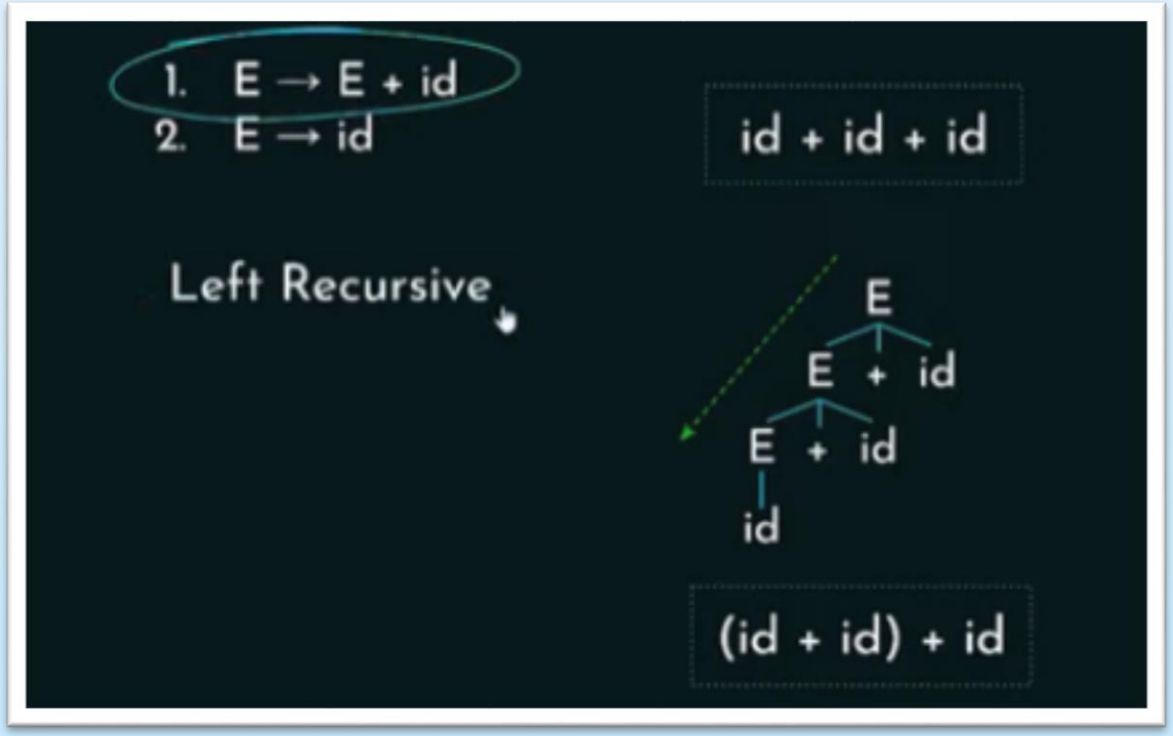


Ambiguity

- Ambiguity Problems:
  - Precedence Property Violation
  - Associatively Property Violation (Solution)
    - Left Recursion = NT on the left of P = Left association
    - Right Recursion = NT on the right of P = Right association

Ambiguity

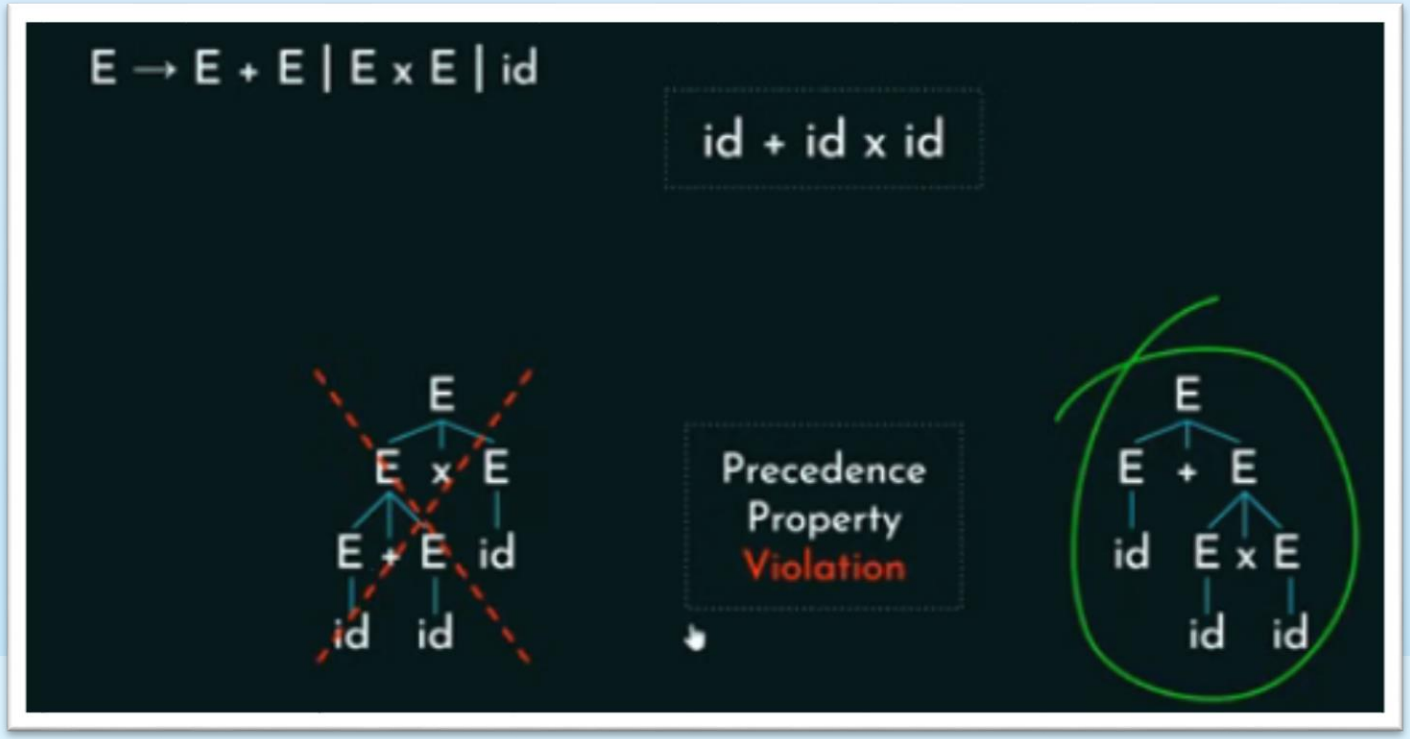
- Ambiguity Problems:
  - Precedence Property Violation
  - Associatively Property Violation



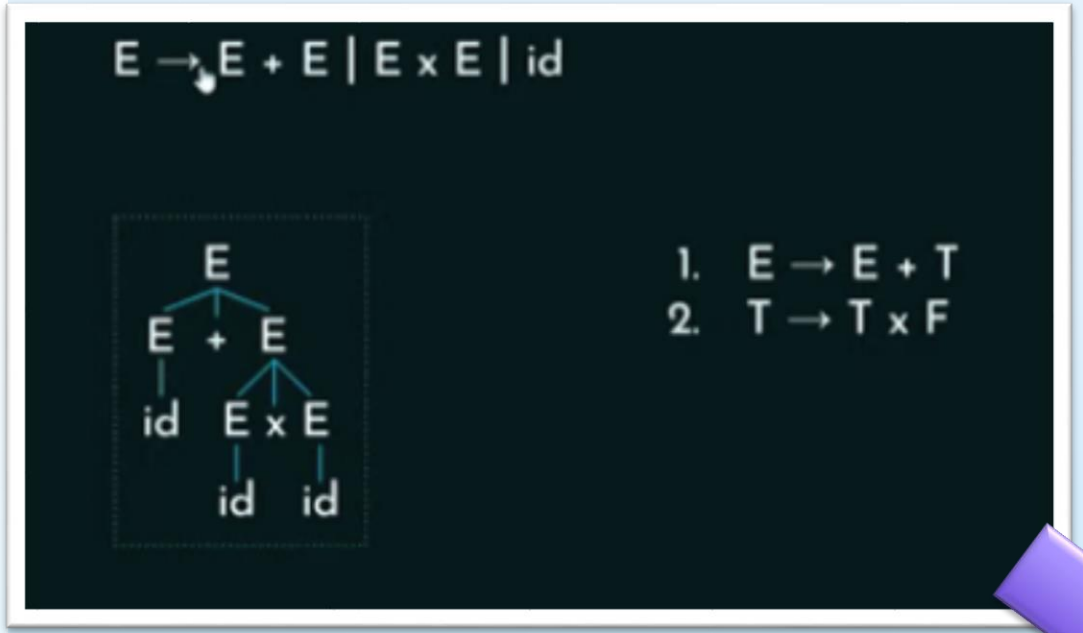
Ambiguity

- Ambiguity Problems:
  - Precedence Property Violation (Solution)
  - Associatively Property Violation

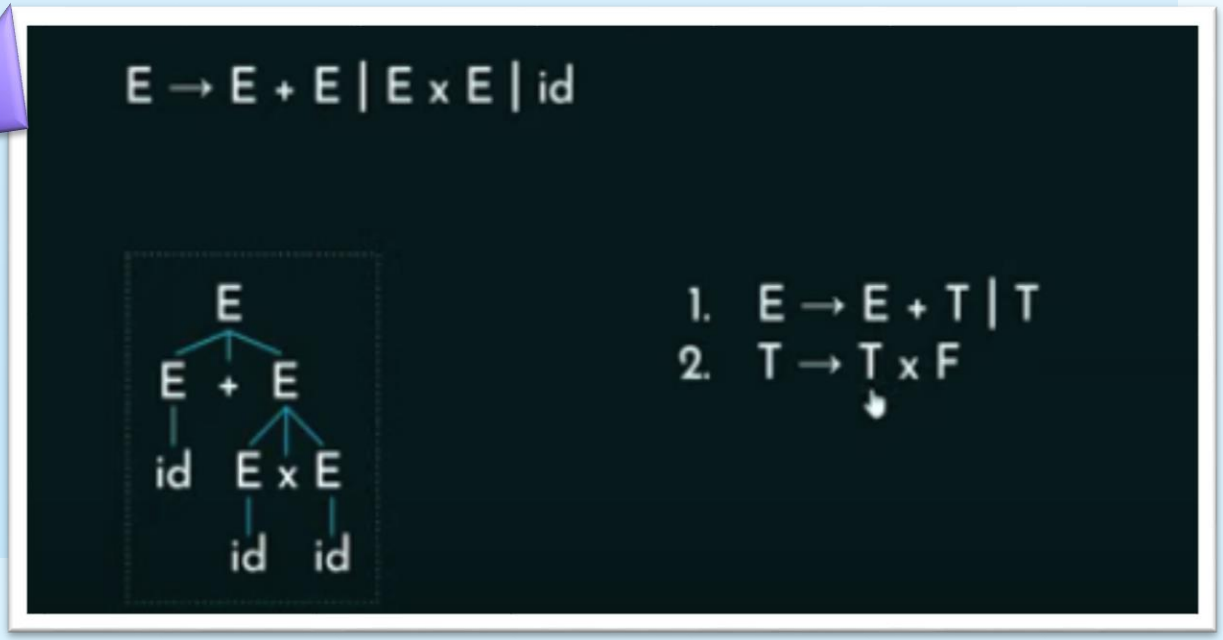
- Ambiguity Problems:
  - Precedence Property Violation
    - Tree is traversed from bottom to top and from left to right. The correct results are produced when the operator of higher precedence is in the lower level.
  - Associatively Property Violation



Ambiguity



Precedence Property



Ambiguity

1. $E \rightarrow E + T \mid T$
2. $T \rightarrow T \times F \mid F$
3. $F \rightarrow id$

Precedence Property resolved by defining levels

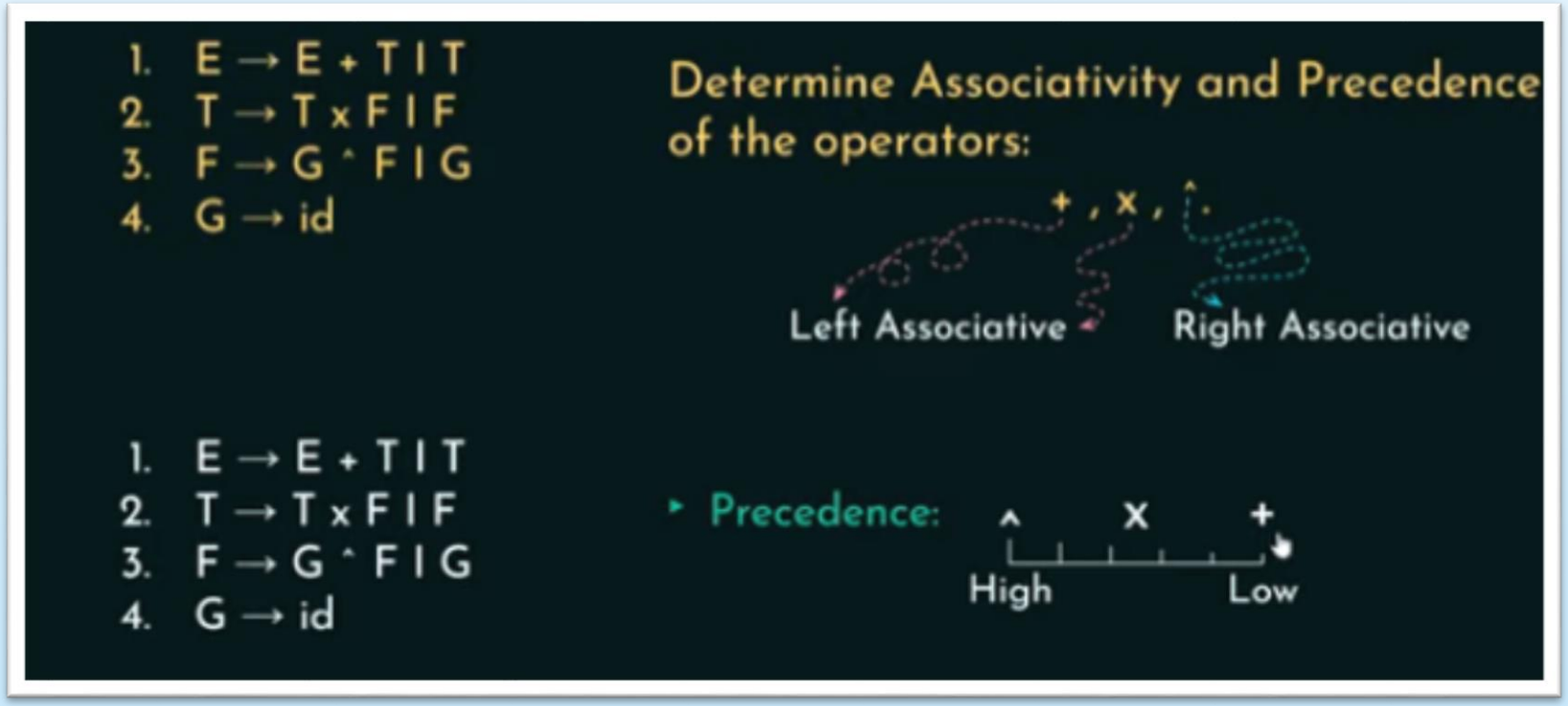
Example


1. E → E + T I T
2. T → T x F I F
3. F → G ^ F I G
4. G → id
Determine Associativity and Precedence of the operators:
+ , x , ^.
Left Associative
1. E → E + T I T

Example


1. E → E + T | T
2. T → T x F | F
3. F → G ^ F | G
4. G → id
Determine Associativity and Precedence of the operators:
+ , x , ^
Left Associative
Right Associative
1. E → E + T | T
2. T → T x F | F
3. F → G ^ F | G

Example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Determine Associativity and Precedence of the operators:
+ , x , ^
Left Associative
Right Associative
1. E → E + T | T
2. T → T x F | F
3. F → G ^ F | G
4. G → id
Precedence:
^ x +
High
Low

Example

Consider the following ambiguous CFG on boolean expressions:

bExp → bExp AND bExp | bExp OR bExp | NOT bExp | True | False

The precedence of the boolean operators are NOT, AND, OR (high to low). AND, OR have left to right associativity.

1. bExp → bExp OR bExp
2. bExp → bExp AND bExp
3. bExp → NOT bExp
4. bExp → True
5. bExp → False

Example

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bExp → bExp AND bExp | bExp OR bExp | NOT bExp | True | False

The precedence of the boolean operators are NOT, AND, OR (high to low). AND, OR have left to right associativity.

1. bExp → bExp OR bExp1
2. bExp1 → bExp1 AND bExp2
3. bExp2 → NOT bExp2
4. bExp2 → True
5. bExp2 → False

Example

Consider the following ambiguous CFG on boolean expressions:

bExp → bExp AND bExp | bExp OR bExp | NOT bExp | True | False

The precedence of the boolean operators are NOT, AND, OR (high to low). AND, OR have left to right associativity.

1. bExp → bExp OR bExp1 | bExp1
2. bExp1 → bExp1 AND bExp2 | bExp2
3. bExp2 → NOT bExp2
4. bExp2 → True
5. bExp2 → False

Example

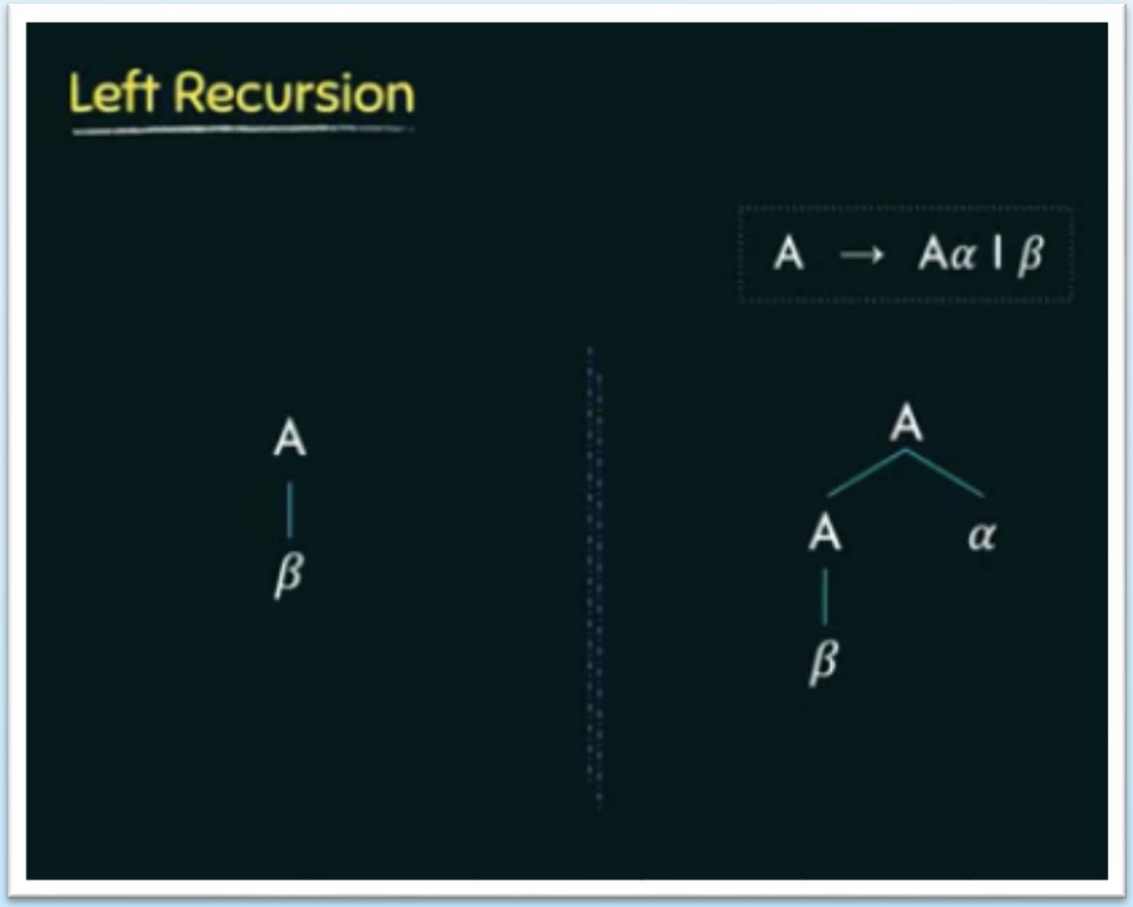
Consider the following ambiguous CFG on boolean expressions:

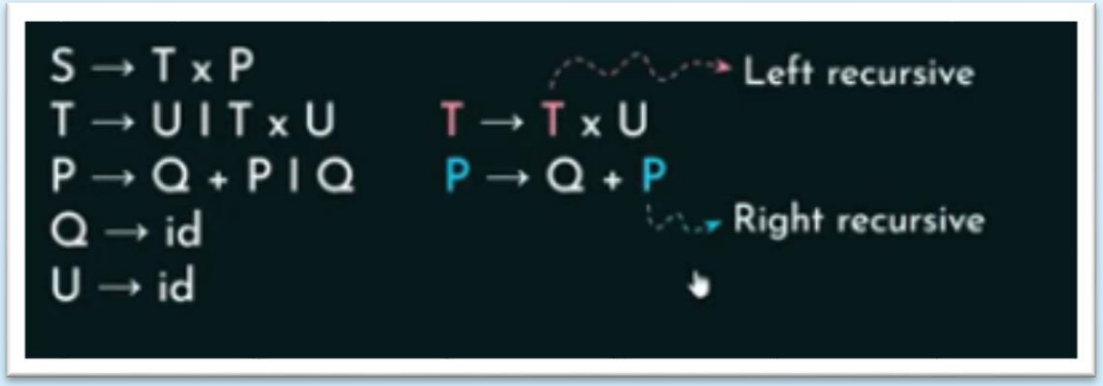
bExp → bExp AND bExp | bExp OR bExp | NOT bExp | True | False

The precedence of the boolean operators are NOT, AND, OR (high to low). AND, OR have left to right associativity.

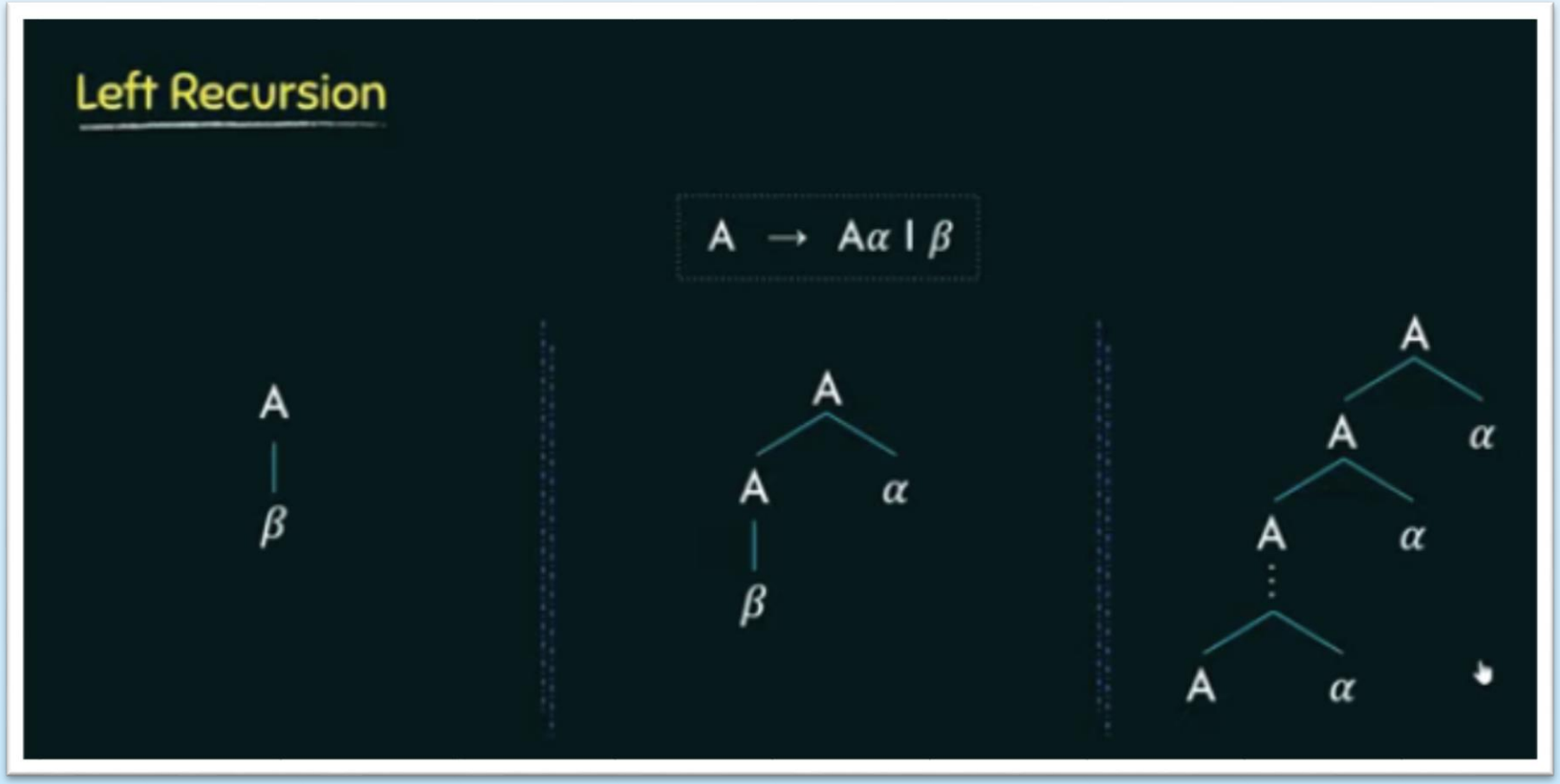
1. bExp → bExp OR bExp1 | bExp1
2. bExp1 → bExp1 AND bExp2 | bExp2
3. bExp2 → NOT bExp2 | True | False

Left and Right Recursions





Left and Right Recursions



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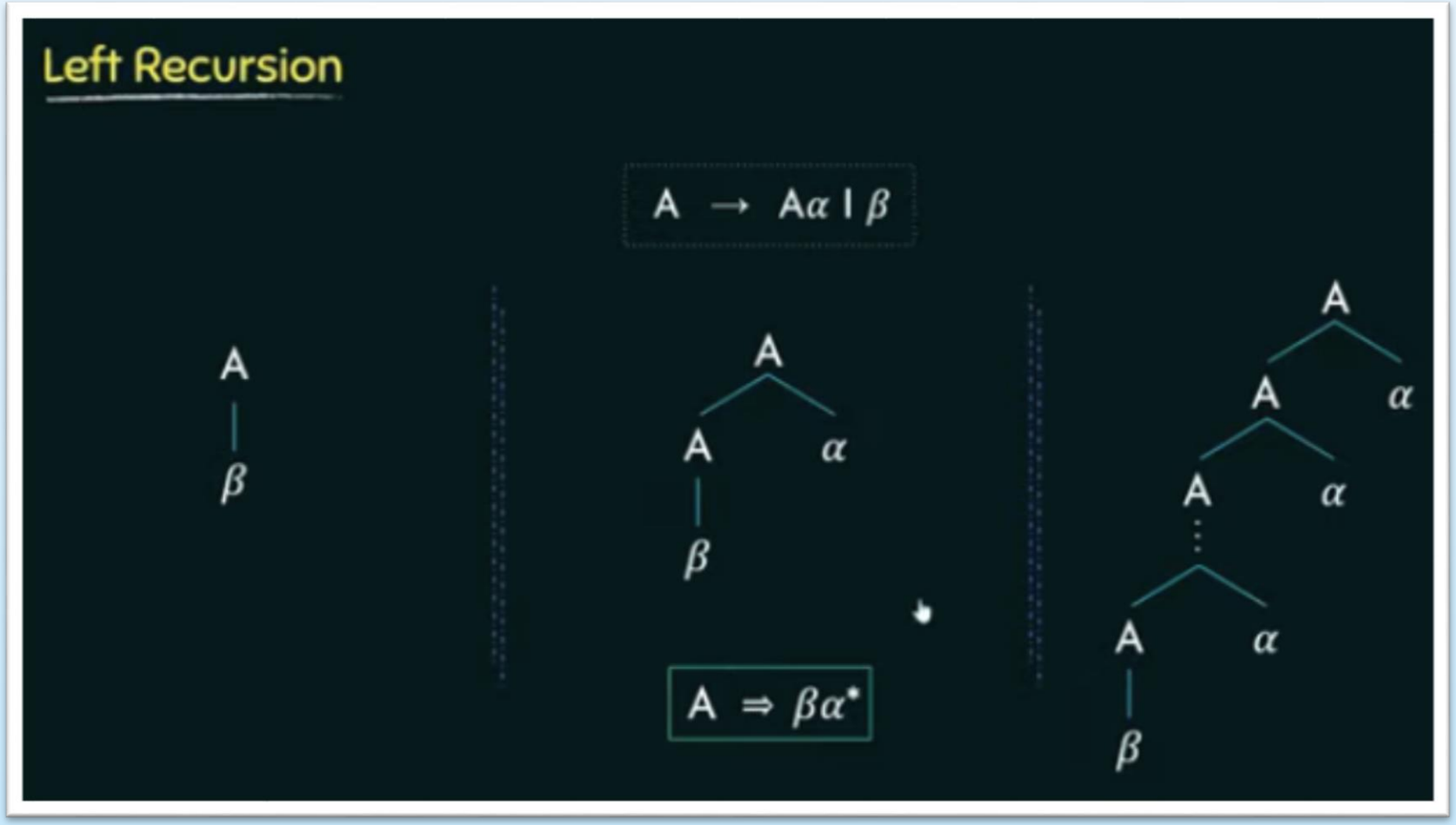

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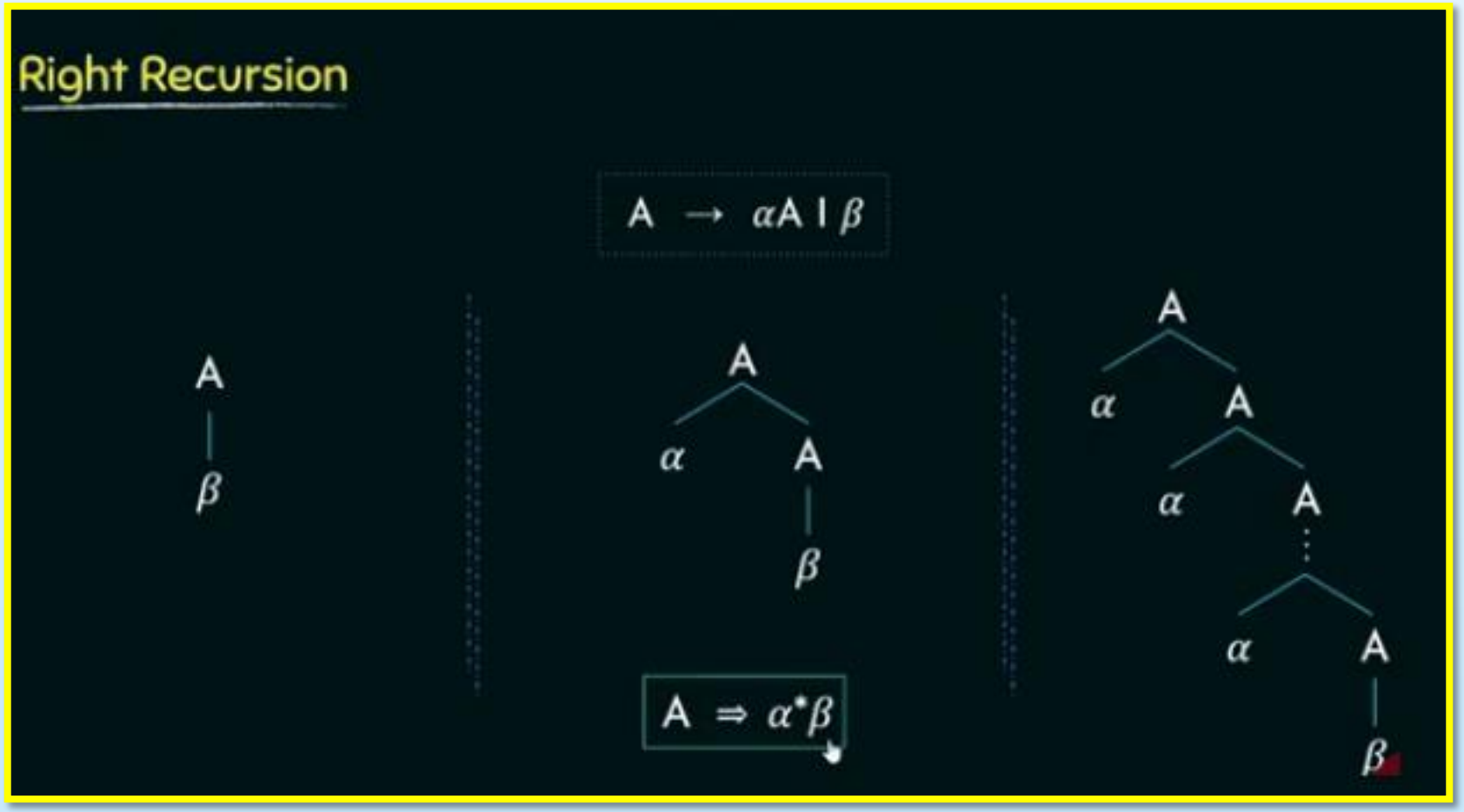
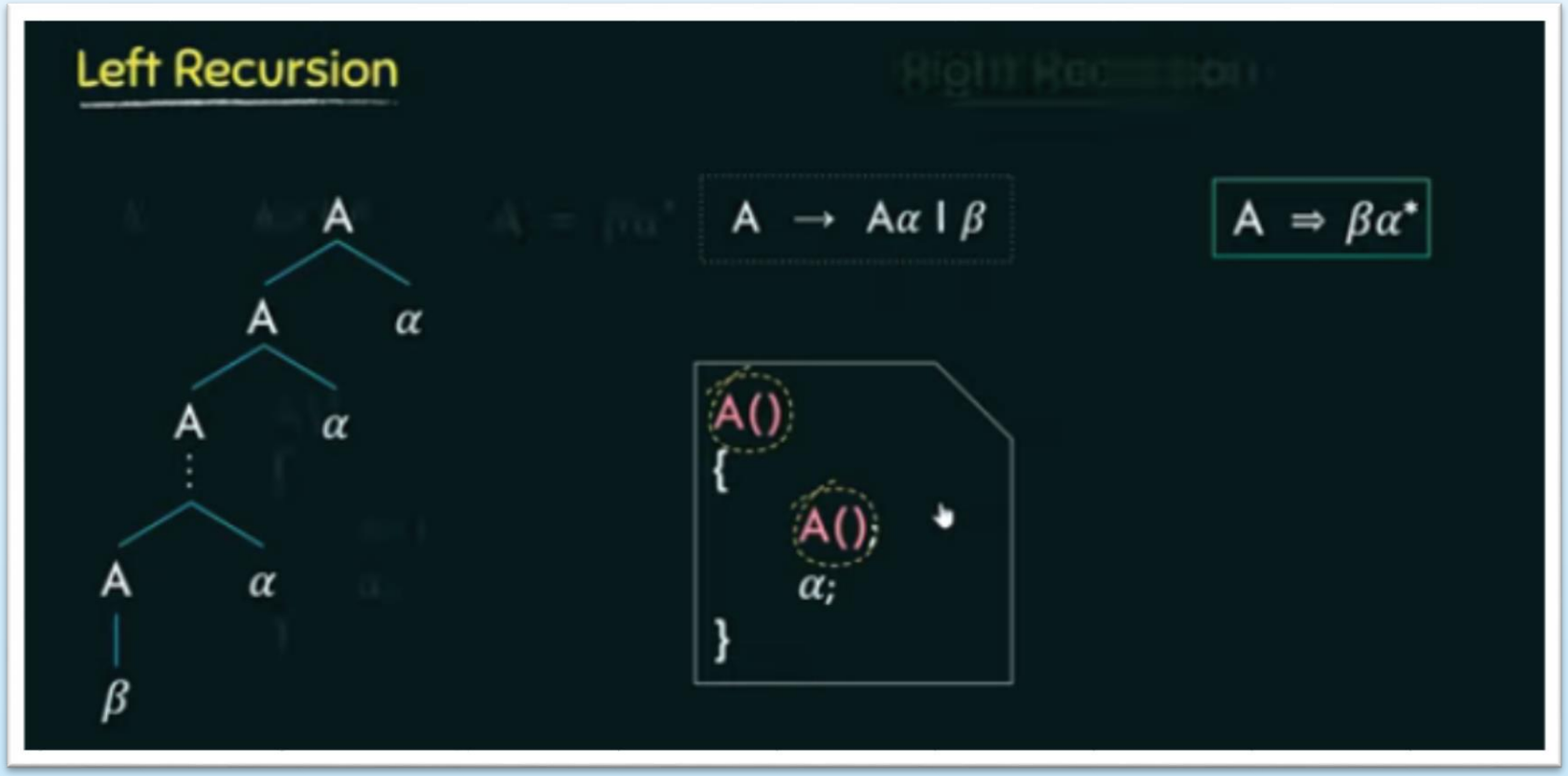
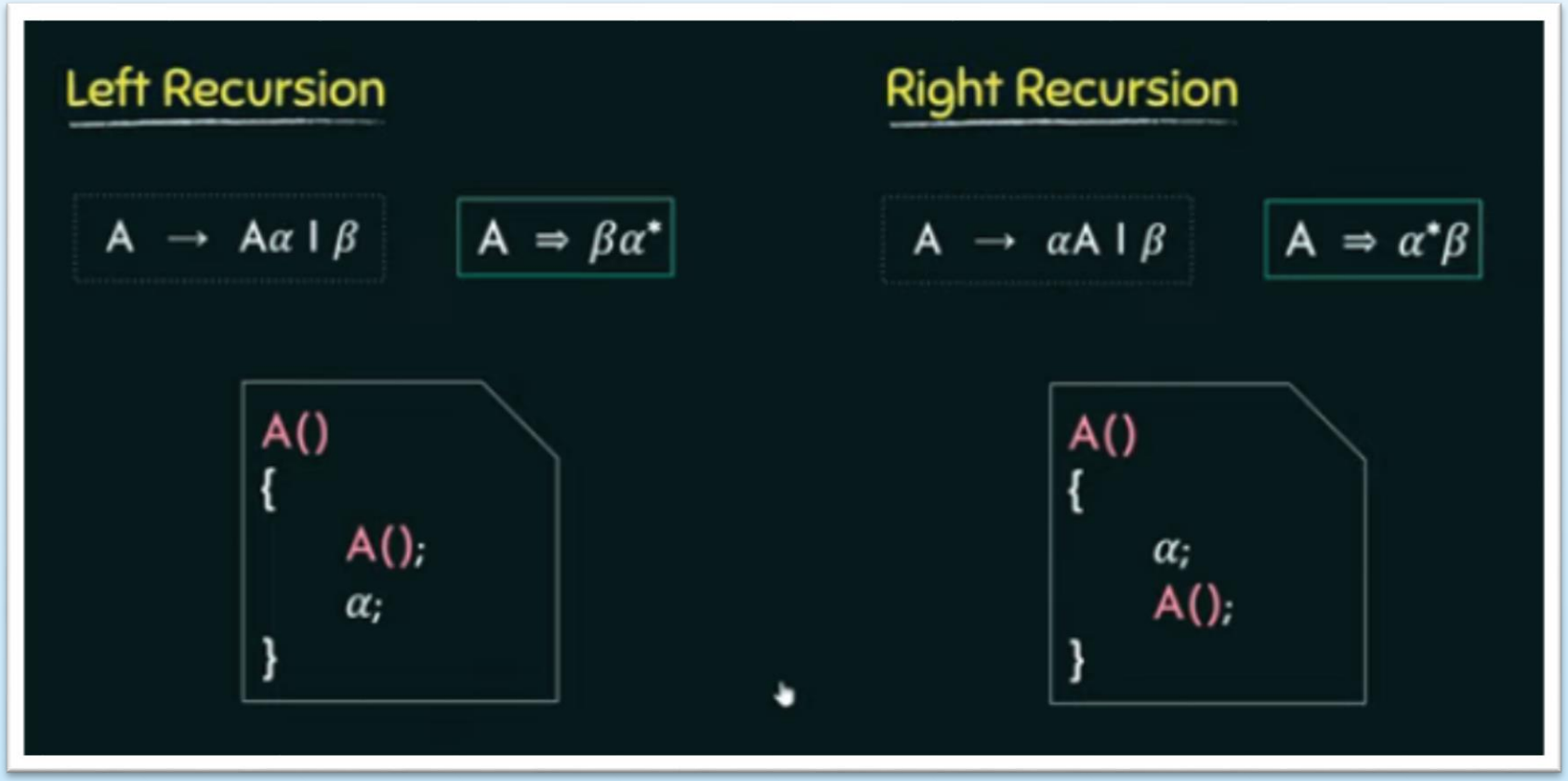


Left and Right Recursions

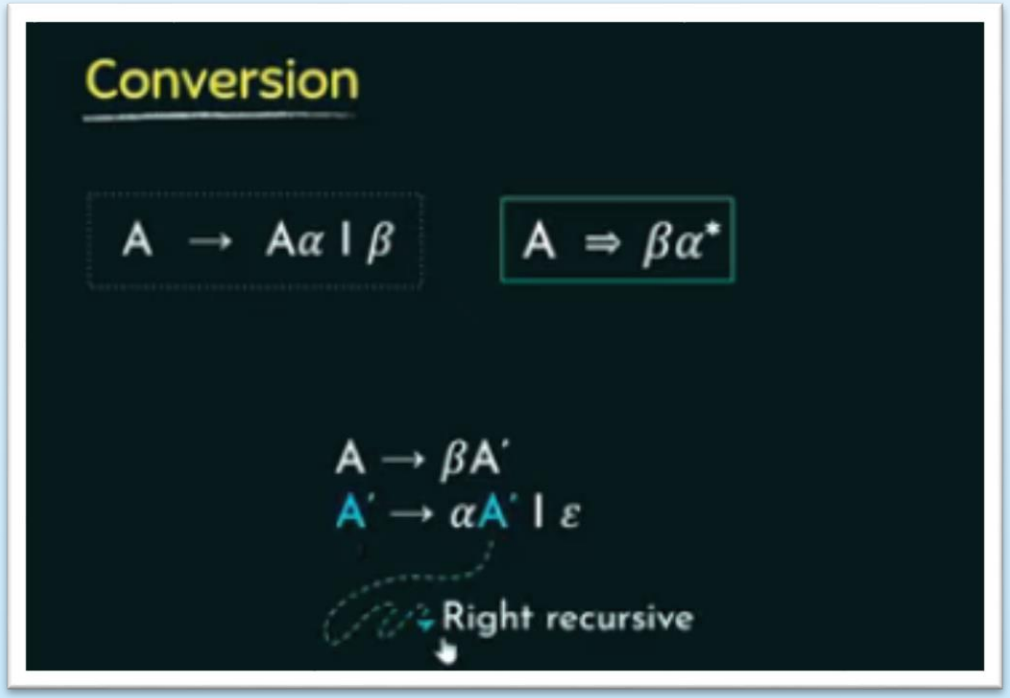


Problem with Left recursion is the infinite loop and confuses the top-down parser

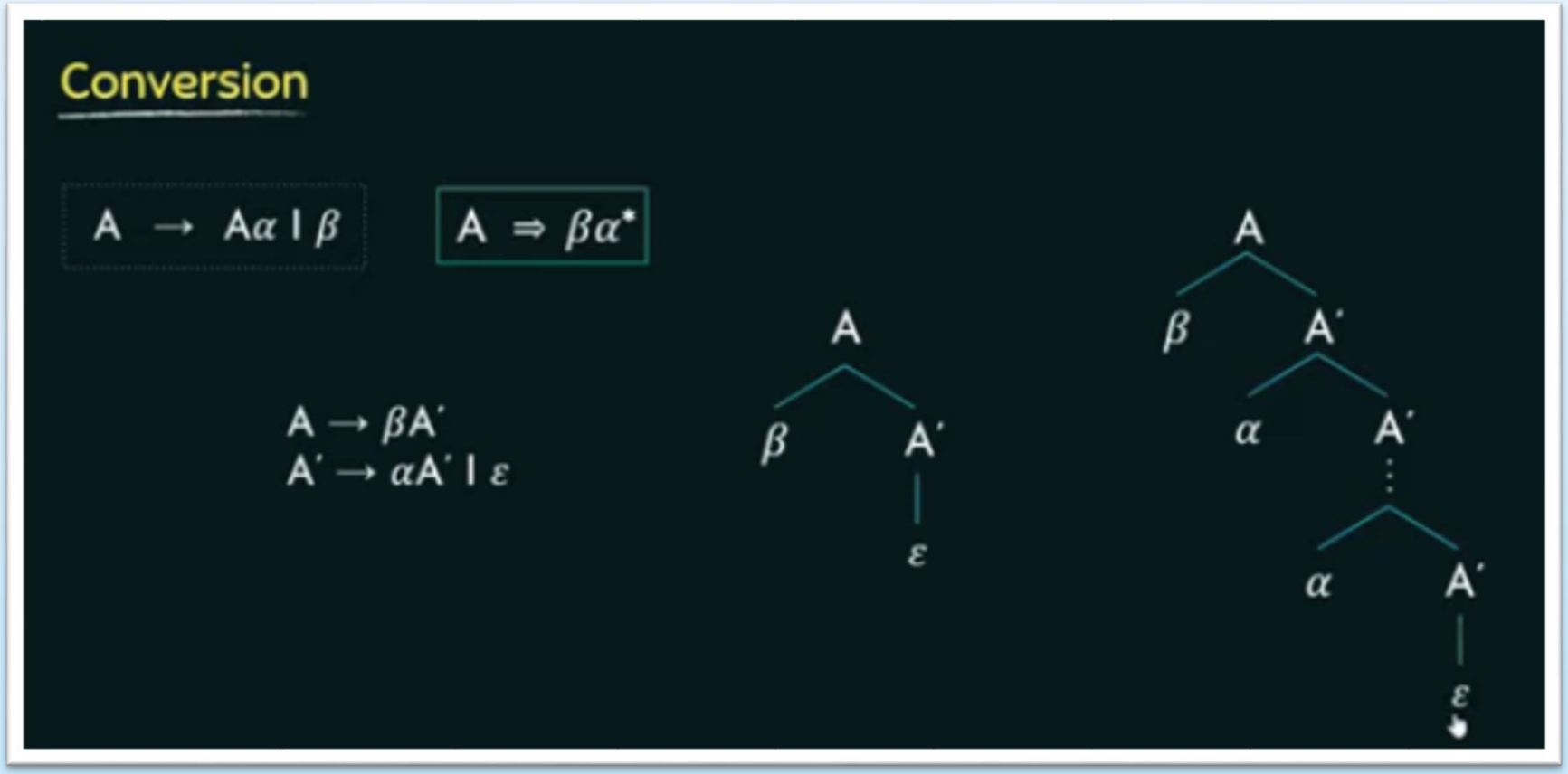
Left and Right Recursions



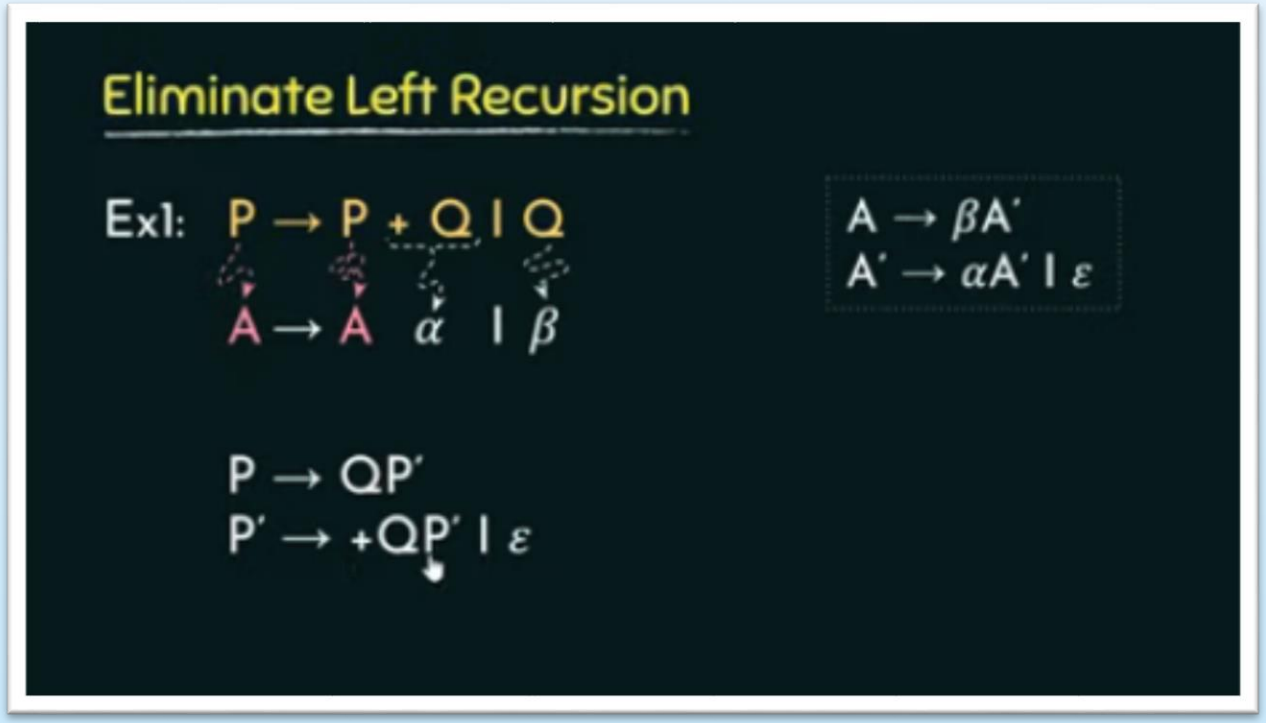
Left Recursion (solution: Right Recursion)



Left Recursion (solution: Right Recursion)



Left Recursion (solution: Right Recursion)



Parse Tree Construction Approaches

- Two distinct and opposite approaches for constructing the tree:
  - Top-Down Parsers begin with the root and grow the tree toward the leaves. At each step, a top-down parser selects a node for some nonterminal on the lower fringe of the partially built tree and extends it with a subtree that represents the right-hand side of a production that rewrites the nonterminal.
  - Bottom-Up Parsers begin with the leaves and grow the tree toward the root. At each step, a bottom-up parser finds a substring of the partially built parse tree's upper fringe that matches the right-hand side of some production; it then builds a node for the rule's left-hand side and connects it into the tree.

Thanks!